

Practical Activity Groups — PAG P1 to PAG P8

OCR GCSE Physics A (J249) — Practical Activity Group summaries

OCR splits the required apparatus and techniques into eight **Practical Activity Groups (PAGs)**, P1 to P8, and you must complete at least eight practical activities that cover them. The specification gives one example activity for each PAG (your school may use different ones); each is summarised below with an apparatus diagram and key expected results. You may be asked about apparatus, method, variables, or how to analyse the results in the written papers — around 15% of the marks test practical skills.

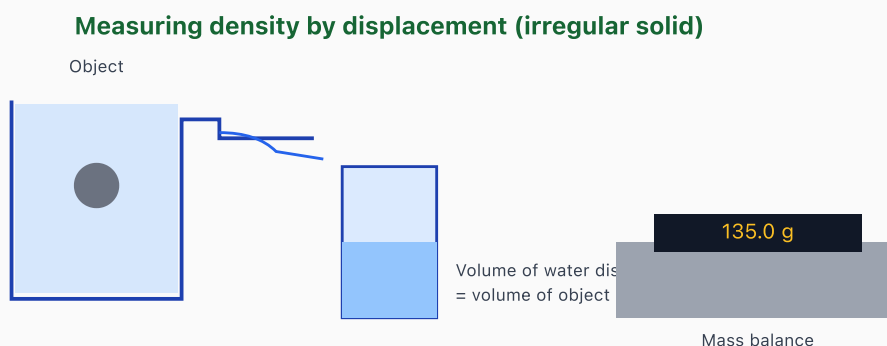
PAG P1 — Materials: Density of Solid Objects and Liquids

Aim: determine the density of regular solids, irregular solids, and a liquid.

Independent variable: object / liquid used.

Dependent variable: density (calculated).

Control variable: temperature (the density of a liquid changes slightly with temperature), use same balance.



Method: Regular solid: measure mass with a balance; measure dimensions with a ruler / vernier calliper; calculate V . Then $\rho = m / V$.

Irregular solid: measure mass. Use a eureka can full to the spout — lower the object; collect displaced water in a measuring cylinder. Volume of water = volume of object.

Liquid: place an empty measuring cylinder on a balance; tare to zero; pour in a known volume and record the mass.

⚠ Safety: Wear eye protection. Handle glassware carefully — wet benches cause slipping. Dry any spilled water.

Exam Tip: Common error: reading the volume at the top of the meniscus rather than the bottom. Always read to the bottom of the meniscus at eye level.

PAG P2 — Forces: Effect of Forces on Springs (Hooke's Law)

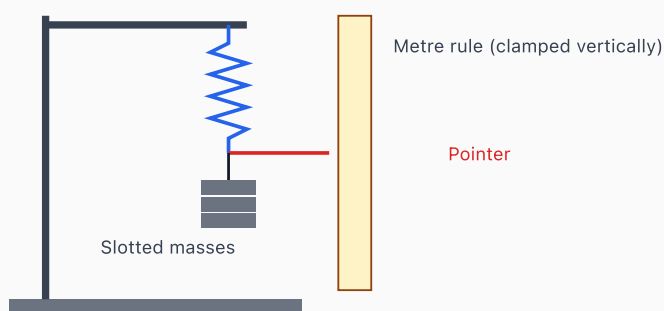
Aim: investigate how the extension of a spring depends on the force applied, find the spring constant, and use the force–extension graph to find the work done in stretching it.

Independent variable: force (weight of slotted masses, $W = mg$).

Dependent variable: extension of the spring.

Control variables: same spring, same starting (unloaded) length, readings taken from the same point on the pointer.

Spring, pointer and metre rule on a clamp stand



- Method:**
1. Clamp the spring from a clamp stand and fix a metre rule vertically beside it, with a pointer attached to the bottom of the spring.
 2. Record the pointer reading with no load (or just the hanger).
 3. Add one slotted mass at a time (e.g. 100 g = about 1 N). After each, let the spring settle and record the new pointer reading.
 4. Extension = new reading – original reading.
 5. Continue for at least 6 loads, then remove the masses one by one to check whether the spring returns to its original length (elastic limit not exceeded).
 6. Plot force (y-axis) against extension (x-axis).

$$F = k x \quad \text{energy stored} = \frac{1}{2} k x^2 \quad (\text{for the linear region})$$

- A straight line through the origin shows extension is proportional to force (Hooke's law); the gradient is the spring constant k (in N/m — convert cm to m).
- Beyond the limit of proportionality the graph curves.
- The area under the force–extension graph is the work done in stretching the spring (energy stored).

⚠ Safety: Wear eye protection in case the spring snaps or flies off. Put a tray or foam under the masses and keep feet clear. Clamp the stand to the bench so it cannot topple.

Exam Tip: Read the pointer with your eye level with it (or use a set square) to avoid parallax error. Extension is the *increase* in length, not the total length.

PAG P3 — Motion: Acceleration of a Trolley Down a Ramp (Newton's Second Law)

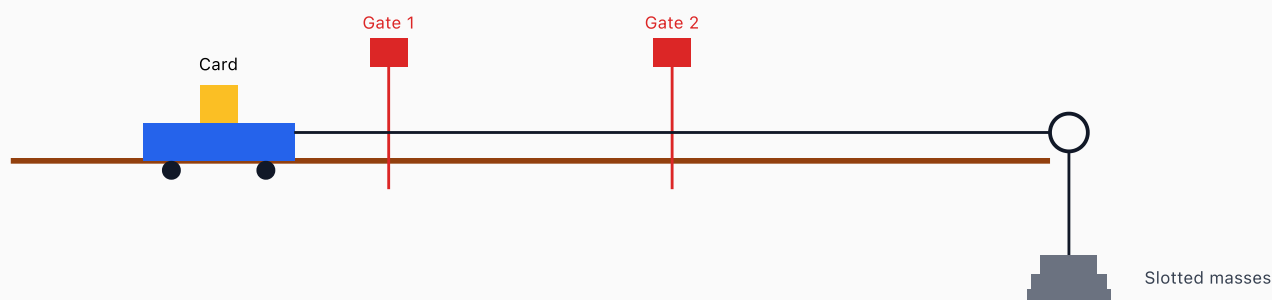
Aim: investigate how the acceleration of a trolley depends on (a) accelerating force and (b) total mass.

Independent variable: force (via slotted masses on a hanger) or total mass of the trolley system.

Dependent variable: acceleration.

Control variables: surface used, same trolley, same light gate spacing.

Trolley, light gates, pulley and mass hanger



Data logger records time to travel between gates; $a = (v - u) / t$.

- Method:**
1. Set up: ramp (use a slight tilt only to compensate for friction so trolley moves at steady velocity when given a gentle push).
 2. Place one light gate near the top of the ramp and a second light gate a measured distance away.
 3. Attach a card to the trolley so it can interrupt the light gate beams.
 4. Add masses to the hanger to accelerate the trolley — record the mass (so $F = mg$ is known).
 5. Release the trolley, record the times / velocities from each gate.
 6. Calculate acceleration $a = (v - u) / t$.
 7. **Part A:** keep trolley mass constant, vary hanging mass (accelerating force). Plot a against $F \rightarrow$ straight line through origin.
 8. **Part B:** keep F constant, add mass to the trolley. Plot a against $1/m \rightarrow$ straight line through origin.

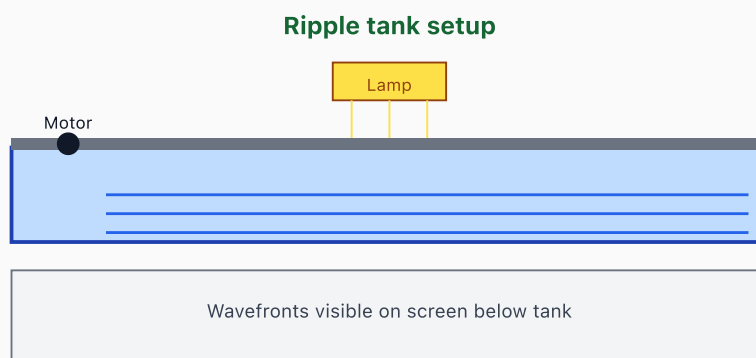
Expect: $a \propto F$ (at constant m) and $a \propto 1/m$ (at constant F). Consistent with $F = m a$.

⚠ Safety: Clamp the pulley securely. Place a folded towel / tray below the hanger to catch falling masses.

PAG P4 — Measuring Waves: Speed, Frequency and Wavelength (Ripple Tank and Stretched String)

Aim: measure the speed, frequency and wavelength of water waves in a ripple tank (and of transverse waves on a stretched string), and judge how suitable the apparatus is for each measurement.

Method A — Water waves in a ripple tank



Method: Use a vibrating bar driven by a motor. Light above casts a shadow of the wavefronts on a screen below.

Frequency f : count the waves passing a point on the screen in 10 s and divide by 10 (or use a stroboscope to freeze the pattern).

Wavelength λ : place a ruler next to the shadow (or photograph it), measure across several wavefronts and divide by the number of wavelengths.

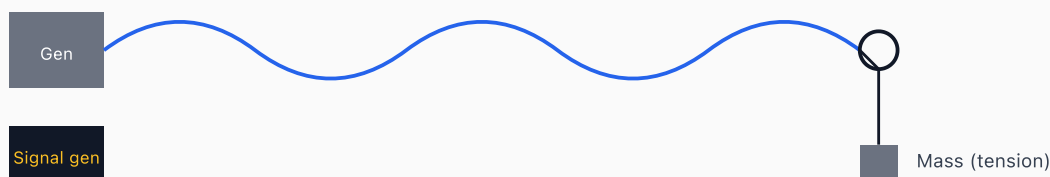
Speed: $v = f \times \lambda$; check it by timing one wavefront over a measured distance ($v = \text{distance} / \text{time}$).

Suitability: ripples are slow and low-frequency, so a ruler, stopwatch and camera are

suitable; faster waves (e.g. sound in a solid rod) need electronic equipment such as a microphone and oscilloscope to measure frequency.

Method B — Transverse waves on a string

Vibration generator, pulley, masses



Method: Fix a known length of string between a vibration generator and a pulley with a mass providing tension. Use a signal generator to set the frequency. Increase f until a stable standing wave forms (e.g. two loops). From $\lambda = 2L/n$ (n = number of loops) and given f , compute v .

⚠ Safety: Keep fingers clear of the vibration generator. Wear eye protection if strings might snap.

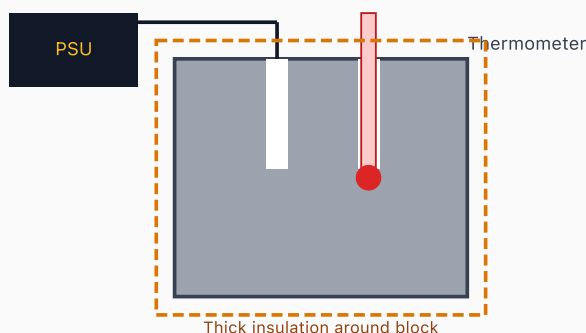
PAG P5 — Energy: Specific Heat Capacity of a Metal

Aim: determine the specific heat capacity (c) of a metal block.

Independent variable: time (as block heats).

Dependent variable: temperature rise of block.

Control variables: block material, voltage / current through heater, same insulation.

Metal block, immersion heater and thermometer

Measure mass (m) first
Record V & I of heater
Time t until $\Delta\theta$ noted

- Method:**
1. Weigh the block (mass m). Insulate it well (fibreglass / polystyrene).
 2. Insert heater in one hole; thermometer in the other with a drop of oil to improve thermal contact.
 3. Record initial temperature θ_0 .
 4. Switch on heater, record V and I. Start timer.
 5. Record temperature every 30 s for 10 minutes.
 6. Plot temperature vs time. Take $\Delta\theta$.
 7. Energy supplied $E = V \times I \times t$.
 8. $c = E / (m \times \Delta\theta)$.

$$\text{Specific heat capacity: } c = V I t / (m \Delta\theta)$$

Exam Tip: The measured c is usually *higher* than the true value because some energy is lost as heat to the surroundings — this is why good insulation is needed.

! Safety: The block and heater get very hot — do not touch for several minutes after switch-off. Mains-operated PSUs should be set to the correct voltage.

PAG P6 — Circuit Components: I-V Characteristics and Resistance

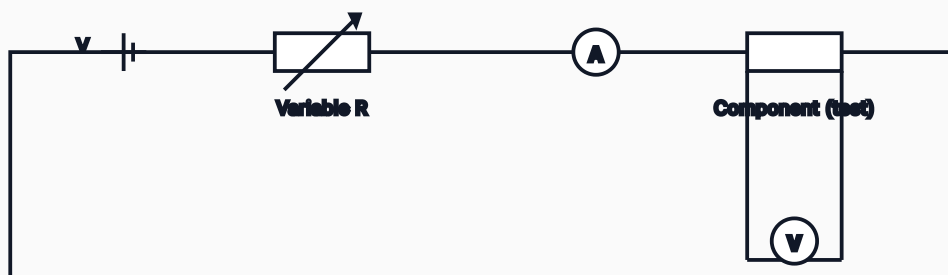
Aim: investigate how current through a component varies with pd for a fixed resistor, a filament lamp, and a diode.

Independent variable: pd across component.

Dependent variable: current through component.

Control variables: ambient temperature, same component throughout.

Circuit: variable resistor controls pd; ammeter in series, voltmeter in parallel



- Method:**
1. Build the circuit with the component under test in the main loop, ammeter in series, voltmeter in parallel with the component.
 2. Adjust the variable resistor from minimum $\rightarrow$ maximum pd across the component, recording pairs (V, I) at ~ 0.5 V steps.
 3. **Reverse the cell connections** and repeat to get negative V values (important for the diode).
 4. Plot I (y-axis) against V (x-axis).

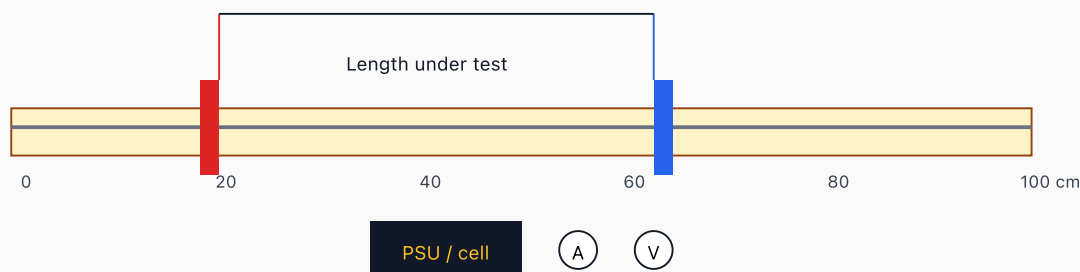
Component	Shape of I–V graph	Interpretation
Fixed resistor at constant T	Straight line through origin	Ohmic — R is constant
Filament lamp	S-shape through origin, flatter at extremes	Non-ohmic; resistance rises as filament heats
Diode	Zero current in reverse; rapid rise above threshold (~ 0.6 V) in forward	Only conducts one way above threshold

⚠ Safety: The filament lamp gets hot. Keep pd within the component's rating or it will burn out. Disconnect between readings if the lamp glows bright.

Further P6 activities (suggested in the specification): how the resistance of a wire depends on its length, and how thermistor / LDR resistance varies with conditions.

Wire length

Resistance wire along metre ruler, crocodile clips, circuit



Method: 1. Attach one crocodile clip to 0 cm; the second clip moves along to select a length L .
2. At each length, close the switch briefly, record V and I . *Open the switch between readings* — otherwise the wire heats up and R changes.
3. Calculate $R = V / I$. Plot R against L — straight line through origin. Gradient gives resistance per metre.

Thermistor (vs temperature)

Method: Immerse the thermistor in a water bath. Heat slowly. Record R (from V / I) at each temperature. Plot R against T — curve *decreases* rapidly as temperature rises (negative temperature coefficient).

LDR (vs light intensity)

Method: Place the LDR a measured distance from a lamp. Vary the distance (or light intensity using filters). Record R at each intensity. R decreases as light level rises.

⚠ Safety: Keep current low ($< 0.5\text{ A}$ for nichrome wire). Do not leave the circuit on continuously — the wire gets hot and can burn. Use insulated crocodile clips.

Exam Tip: Measure the **diameter** of the wire with a micrometer screw gauge in several places, take an average. A thinner wire has higher resistance for the same length.

PAG P7 — Series and Parallel Circuits: Brightness of Bulbs

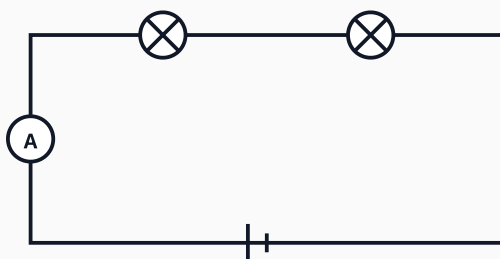
Aim: build series and parallel circuits from circuit diagrams, compare the brightness of identical bulbs, and measure the current and potential difference in each arrangement.

Independent variable: circuit arrangement (series or parallel) and the number of bulbs.

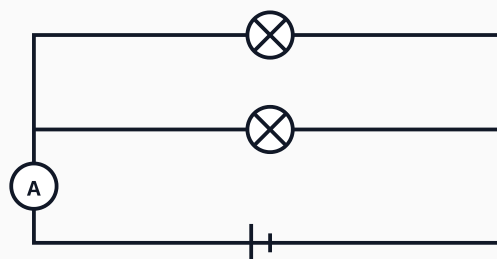
Dependent variables: brightness of each bulb; current and p.d. readings.

Control variables: same supply p.d., identical bulbs, same meters.

Two identical bulbs: series (left) and parallel (right)



Bulbs share the supply p.d. → dimmer



Each bulb gets the full supply p.d. → brighter

Method: 1. Build a circuit with one bulb, a switch and an ammeter. Note its brightness and record the current and the p.d. across the bulb.
2. Add a second identical bulb **in series**. Compare the brightness; measure the

current at different points and the p.d. across each bulb.

3. Rebuild with the two bulbs **in parallel**. Measure the current in each branch and in the main circuit, and the p.d. across each bulb.

4. Repeat with three bulbs if time allows, and check each circuit against its diagram before switching on.

	Series	Parallel
Current	Same at every point	Branch currents add up to the total current
Potential difference	Supply p.d. is shared between the bulbs	Each branch has the full supply p.d.
Total resistance	Sum of the resistances — increases as bulbs are added	Less than the smallest branch resistance — decreases as bulbs are added
Brightness	Bulbs get dimmer as more are added	Each bulb stays about as bright as a single bulb

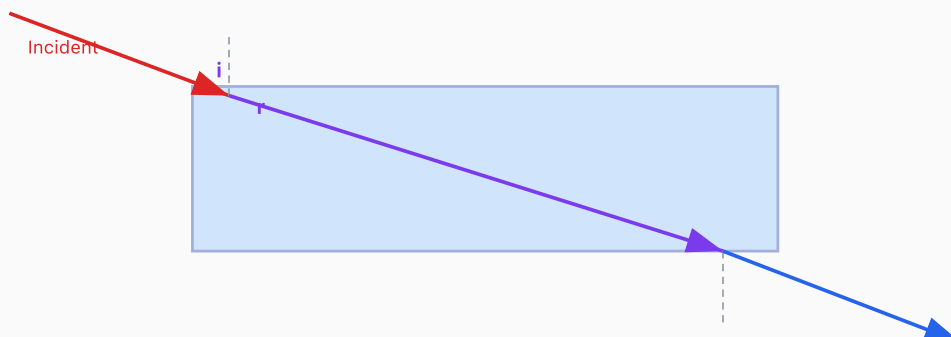
⚠ Safety: Use a low-voltage supply. Bulbs get hot — switch off between readings and let them cool before handling.

Exam Tip: Brightness depends on the power of the bulb ($P = IV$). In parallel each bulb has the full p.d. and a larger current, so it transfers more energy per second than a bulb in series.

PAG P8 — Interactions of Waves: Reflection and Refraction of Light

Aim: investigate how a light ray changes direction when it is reflected by a plane mirror and when it passes through a glass block or prism.

Ray box and glass block on plain paper



Method: 1. Place the glass block on plain paper; draw around it.
2. Shine a narrow ray from a ray box at an angle of incidence i onto one long side.
3. Mark where the ray enters and leaves the block.
4. Remove the block; join up the dots to reconstruct the path inside the block.
5. Draw normals at the entry and exit points.
6. Measure i and r with a protractor.
7. Repeat for several angles of incidence (e.g. 20° , 30° , 40° , 50° , 60°).
8. Plot r against i (not a straight line) or $\sin r$ against $\sin i$ (straight through origin — gradient = $1 / \text{refractive index}$).

Reflection from a plane mirror

Method: 1. Stand a plane mirror upright on a line drawn across a sheet of plain paper and draw a normal at 90° to the line.
2. Aim a narrow ray from the ray box at the point where the normal meets the mirror.
3. Mark the incident and reflected rays with crosses; remove the mirror and join them up.
4. Measure the angle of incidence and the angle of reflection from the normal.
Repeat for several angles — they are equal.

Prisms

Method: Repeat the refraction method with a triangular prism: the ray bends towards the normal on entering and away from the normal on leaving, so it is deviated towards the base of the prism. With white light, the different colours refract by different amounts and a spectrum is seen.

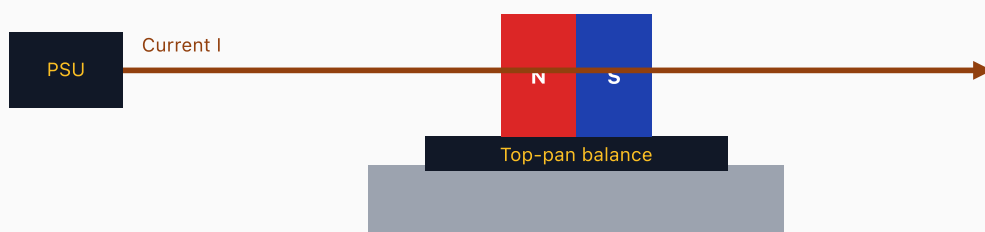
Exam Tip: Use a **thin** pencil to mark ray positions. Use at least three dots along each ray so you can draw straight lines accurately. Measure angles *from the normal*, not from the block surface.

Not a PAG example — Force on a Current-Carrying Wire in a Magnetic Field

This is not one of the eight PAG example activities in the J249 specification, but it is useful supporting practice for the motor effect and $F = B I l$ (topic P4.2).

Aim: investigate the force on a current-carrying conductor in a magnetic field.

Current-carrying wire between magnet poles on top-pan balance



Method: 1. Clamp a pair of opposite magnetic poles so the field passes horizontally between them.

2. Place the magnets on a top-pan balance; tare to 0.
3. Pass a wire horizontally between the poles (perpendicular to the field).
4. Switch on the current; note the reading on the balance.
5. A change in reading shows the force on the wire (Newton's 3rd law: wire pushes on the magnets, magnets push wire up/down).
6. Vary the current I ; plot mass change against I . Line should be straight through origin — $F \propto I$.
7. Reverse the current $\rightarrow$ force reverses (reading swaps sign).

$F = B I L$ — force is proportional to current (this practical checks that).

 **Safety:** Keep current values low (≤ 5 A) and limit duration — wires get hot. Use a thick wire to reduce resistance.

Practical Skills — Generic Points (Assessed on Any PAG)

Skill	What it means
Accuracy	How close a reading is to the true value. Reduced by calibrated instruments and correct technique.
Precision	How small the scatter in repeated readings is. Improved by repeats and suitable apparatus resolution.
Resolution	The smallest change an instrument can measure. Ruler: 1 mm. Micrometer: 0.01 mm.
Random error	Unpredictable variation. Reduce by averaging repeated readings.
Systematic error	Consistent offset (e.g. zero-error). Reduce by checking zero / calibration.

Anomalous result	A reading that doesn't fit the pattern — identify and repeat; do not include in averages.
Repeatability / Reproducibility	Repeatability: same person, same equipment, same result. Reproducibility: different people/kit give the same result.

Exam Tip: When writing up a practical: state hypothesis → draw labelled diagram → list IV, DV, CVs → detail step-by-step method (so another student could copy it) → include a safety section → describe how data will be analysed.